ITACA CSA: per-event production scaling

Generated from saved timing measurements

2026-11-06

Fresh outputs, after a 10-event warm-up: 7.891 s for 100 events and 18.701 s for 300 events. The incremental cost is 0.0541 s/event. The projected event-processing cost for 8,000 events is 434.9 s (7.25 min), excluding one-time calibration, process loading and presentation. The previous linear/SPICE 8,000-event job took 1733.8 s (28.90 min), or 0.2167 s/event including its fixed overhead.

Events	Processing (s)	Calibration (s)	Nonlinear pads	SPICE
100	7.891	5.209	342	0
300	18.701	10.448	1125	0

The benchmark uses bb0nu at 4 mm / 40 ms. Each run uses a recorded fresh noise seed to force a fresh response identity; absence of filtered response files is asserted before running. Source ion histories and all physical/electronics parameters are unchanged. Both measured runs use the same warmed Julia process. Covariance calibration is computed and timed separately before event processing; presentation is timed and subtracted. Processing includes linear screening, nonlinear evaluation, noise generation, reconstruction, summaries, output writing, publication and consistency checks. The two-point slope removes remaining fixed processing overhead. No SPICE controls or fallbacks were needed. The 8,000-event value is an extrapolation, subject to event-complexity and disk-throughput variation.

The fitted residual fixed cost is 2.485 s. The earlier 28.46 s pilot reused a filtered response and is excluded from this scaling analysis.

Reproduce with `scripts/production/benchmark_ion_csa_scaling.jl`; the selected timing artifact is recorded in `results/ion_csa_nonlinear/scaling/latest.toml`.